

3 (a)	helium nucleus OR contains two protons and two neutrons	B1	[1]
(b)	kinetic energy = $\frac{1}{2}mv^2$ $\frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times v^2 = 1.07 \times 10^{-12}$ $v = 1.8 \times 10^7 \text{ m s}^{-1}$	C1 A1 A0	[2]
(c) (i)	sum of momenta (in any direction) is constant / total momentum is constant in a closed system / no external force	M1 A1	[2]
(ii)	momentum of francium (= 0) = momentum of α + momentum of astatine $204 \times V = 4 \times 1.8 \times 10^7$ $V = 3.5 \times 10^5 \text{ m s}^{-1}$ <i>(nuclei incorrectly identified, 0/3)</i> <i>nuclei correctly identified but incorrect masses, -1 each error)</i>	C1 C1 A1	[3]
(d)	another particle / photon is emitted at an angle to the direction of the α -particle (allow 1 mark for 'Francium nucleus is not stationary')	M1 A1	[2]

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4 (a) (i) when two (or more) waves meet (at a point)

M1